

Day 3

April 14, 2026

Agenda

- Production Game Review
- Production
 - Bench, Line, Cell Factories
 - DCs
- Marketing
 - Bass Model, Innovators, Imitators, Advertising
 - R&D, Pricing
- Financing
 - Retained Earnings, Working Capital, Debt
- Practice Round
 - Supply = Demand
 - Build the Industry
 - Don't go bankrupt

Important Links

Practice Game: <https://i.responsive.net/gl/axp-practice/>

Please log out of any old games that might be open on your computer.

Please throw away any production game only documents from yesterday.

Please always use current documents and links.

They are posted on Canvas.

Production Game Review

Scoreboard Day 910

RTI: Cash Balance – Debt + Dividends + Returns on Dividends; Cash Position = Cash Balance – Emergency Debt

Winner!

World	Team Name	Return to Investors	Retained Earnings at end of day 910	Cash Position
1	Gotham	\$ <u>846,677</u>	\$ -525,054	\$ <u>846,677</u>
1	Panem	\$ <u>894,015</u>	\$ -953,551	\$ <u>894,015</u>
1	B612	\$ <u>323,414</u>	\$ -964,264	\$ <u>323,414</u>
1	Globex	\$ <u>216,053</u>	\$ -1,014,716	\$ <u>216,053</u>
1	Vulcan	\$ <u>556,291</u>	\$ -1,266,090	\$ <u>556,291</u>
1	Dune	\$ <u>60,809</u>	\$ -1,840,952	\$ <u>60,809</u>
1	Westeros	\$ <u>-210,750</u>	\$ -2,076,101	\$ <u>-210,750</u>
1	Zion	\$ <u>-1,414,474</u>	\$ -2,388,875	\$ <u>-1,414,474</u>

Starting point was -\$342,298

Scoreboard Day 1092

RTI: Cash Balance – Debt + Dividends + Returns on Dividends; Cash Position = Cash Balance – Emergency Debt

Winner!

World	Team Name	Return to Investors	Retained Earnings at end of day 1,092	Cash Position
1	Gotham	\$ <u>1,276,656</u>	\$ -263,337	\$ <u>1,276,656</u>
1	Panem	\$ <u>1,465,880</u>	\$ -795,068	\$ <u>1,465,880</u>
1	B612	\$ <u>437,210</u>	\$ -837,584	\$ <u>437,210</u>
1	Vulcan	\$ <u>794,579</u>	\$ -1,220,104	\$ <u>794,579</u>
1	Globex	\$ <u>-109,457</u>	\$ -1,629,934	\$ <u>-109,457</u>
1	Dune	\$ <u>97,810</u>	\$ -1,919,688	\$ <u>97,810</u>
1	Westeros	\$ <u>-493,807</u>	\$ -2,304,470	\$ <u>-493,807</u>
1	Zion	\$ <u>-1,046,658</u>	\$ -2,792,831	\$ <u>-1,046,658</u>

Starting point was -\$342,298

Gotham – Day 910

Vulcan, Panem, Globex and B612 also became profitable towards the end

	\$	Quarter 10	Quarter 9	Quarter 8	Quarter 7	Quarter 6	Quarter 5	Quarter 4	Quarter 3	Quarter 2
Sales revenue	\$	1,542,000	1,761,450	1,324,300	924,700	630,000	744,100	682,500	543,200	456,400
Cost of goods sold	\$	846,651	927,472	709,913	539,748	361,458	437,447	417,743	332,481	279,352
Gross profit	\$	695,349	833,978	614,387	384,952	268,542	306,653	264,757	210,719	177,048
Advertising expense	\$	9,102	50,496	4,822	640	0	0	0	0	0
Selling expense	\$	321,980	374,540	282,620	198,150	135,000	159,450	146,250	116,400	97,800
Distribution center operating expense	\$	182,000	182,000	182,000	182,000	182,000	182,000	182,000	182,000	182,000
Total selling expense	\$	513,082	607,036	469,442	380,790	317,000	341,450	328,250	298,400	279,800
Research and development expense	\$	0	0	0	0	100,000	409,500	0	0	0
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667
Idle factory expense	\$	0	0	36,087	0	0	0	0	15,932	14,369
Other operating expenses	\$	41,667	41,667	77,754	41,667	141,667	451,167	41,667	57,598	56,036
Operating income	\$	140,600	185,275	67,191	(37,505)	(190,124)	(485,963)	(105,160)	(145,279)	(158,788)
Interest revenue	\$	4,413	3,373	3,964	6,781	9,047	11,938	11,913	12,469	13,554
Net interest and fees revenue (expense)	\$	4,413	3,373	3,964	6,781	9,047	11,938	11,913	12,469	13,554
Total net income before tax	\$	145,013	188,648	71,155	(30,724)	(181,077)	(474,025)	(93,247)	(132,810)	(145,234)
Income tax expense	\$	50,754	66,026	24,904	(10,753)	(63,377)	(165,908)	(32,636)	(46,483)	(50,831)
Net income after tax and before extraordinary items	\$	94,259	122,622	46,251	(19,971)	(117,700)	(308,117)	(60,611)	(86,327)	(94,402)
Net income	\$	94,259	122,622	46,251	(19,971)	(117,700)	(308,117)	(60,611)	(86,327)	(94,402)

Zion– Day 910

Idle time is a profit-killer

	\$	Quarter 10	Quarter 9	Quarter 8	Quarter 7	Quarter 6	Quarter 5	Quarter 4	Quarter 3	Quarter 2
Sales revenue	\$	2,163,360	1,934,750	1,571,750	1,056,600	887,600	789,600	682,500	543,200	456,400
Cost of goods sold	\$	1,482,224	1,212,171	970,737	661,687	532,187	465,696	417,743	332,481	279,352
Gross profit	\$	681,136	722,579	601,013	394,913	355,413	323,904	264,757	210,719	177,048
Selling expense	\$	459,642	411,190	334,700	225,900	190,200	169,200	146,250	116,400	97,800
Distribution center operating expense	\$	182,000	182,000	182,000	182,000	182,000	182,000	182,000	182,000	182,000
Total selling expense	\$	641,642	593,190	516,700	407,900	372,200	351,200	328,250	298,400	279,800
Research and development expense	\$	0	0	20,000	40,000	648,000	520,000	0	0	0
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667	41,667
Idle factory expense	\$	1,827,628	0	10,946	45,254	0	0	0	15,932	14,369
Other operating expenses	\$	1,869,294	41,667	72,612	126,920	689,667	561,667	41,667	57,598	56,036
Operating income	\$	(1,829,801)	87,722	11,701	(139,907)	(706,454)	(588,962)	(105,160)	(145,279)	(158,788)
Interest revenue	\$	3,142	3,369	3,846	5,534	4,895	11,902	11,913	12,469	13,554
Interest expense on emergency loan	\$	15,413	0	0	0	0	0	0	0	0
Net interest and fees revenue (expense)	\$	(12,271)	3,369	3,846	5,534	4,895	11,902	11,913	12,469	13,554
Total net income before tax	\$	(1,842,072)	91,091	15,547	(134,374)	(701,559)	(577,060)	(93,247)	(132,810)	(145,234)
Income tax expense	\$	(644,725)	31,881	5,441	(47,030)	(245,545)	(201,971)	(32,636)	(46,483)	(50,831)
Net income after tax and before extraordinary items	\$	(1,197,347)	59,209	10,106	(87,343)	(456,014)	(375,090)	(60,611)	(86,327)	(94,402)
Net income	\$	(1,197,347)	59,209	10,106	(87,343)	(456,014)	(375,090)	(60,611)	(86,327)	(94,402)

Market Grids – Day 910

Unit sales, average price, revenue

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
MD - Heart	9,817	6,499	11,092	11,115		9,989		10,400	6,500	11,610	77,022
MD - Breast	279	1,094	1,206	57		742		1,853	300	713	6,244
MD - Estrogen		20							502	509	1,031
Law - Narcotic			873	908		107		491		1,015	3,394
Clinic - Fertility											
Total	10,096	7,613	13,171	12,080		10,838		12,744	7,302	13,847	87,691

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Average
MD - Heart	\$ 695	\$ 699	\$ 700	\$ 700		\$ 700		\$ 696	\$ 650	\$ 700	\$ 695
MD - Breast	\$ 1,335	\$ 1,124	\$ 1,220	\$ 1,300		\$ 1,311		\$ 1,120	\$ 1,300	\$ 1,145	\$ 1,185
MD - Estrogen		\$ 850							\$ 680	\$ 750	\$ 718
Law - Narcotic			\$ 1,121	\$ 1,106		\$ 1,250		\$ 1,153		\$ 1,048	\$ 1,104
Clinic - Fertility											
Average	\$ 713	\$ 761	\$ 776	\$ 733		\$ 747		\$ 776	\$ 679	\$ 750	

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
MD - Heart	\$ 6,822,400	\$ 4,545,850	\$ 7,765,320	\$ 7,780,500		\$ 6,992,300		\$ 7,242,350	\$ 4,227,892	\$ 8,127,500	\$ 53,504,112
MD - Breast	\$ 372,425	\$ 1,230,030	\$ 1,471,394	\$ 74,100		\$ 973,050		\$ 2,074,910	\$ 390,000	\$ 816,110	\$ 7,402,019
MD - Estrogen		\$ 17,000							\$ 341,340	\$ 381,750	\$ 740,090
Law - Narcotic			\$ 978,577	\$ 1,003,850		\$ 133,750		\$ 566,050		\$ 1,064,200	\$ 3,746,427
Clinic - Fertility											
Total	\$ 7,194,825	\$ 5,792,880	\$ 10,215,291	\$ 8,858,450		\$ 8,099,100		\$ 9,883,310	\$ 4,959,232	\$ 10,389,560	\$ 65,392,648

Market Grids

Penetration

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
MD - Heart	28.46%	18.84%	32.15%	32.22%		28.95%		30.14%	18.84%	33.65%	22.33%
MD - Breast	1.86%	7.29%	8.04%	<1%		4.95%		12.35%	2%	4.75%	4.16%
MD - Estrogen		<1%							3.35%	3.39%	0.69%
Law - Narcotic			8.73%	9.08%		1.07%		4.91%		10.15%	3.39%
Clinic - Fertility											
Total	8.11%	6.11%	10.58%	9.70%		8.71%		10.24%	5.87%	11.12%	

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
MD - Heart	28%	18%	32%	32%		28%		30%	18%	33%	22.33%
MD - Breast	1%	7%	8%	<1%		4%		12%	2%	4%	4.16%
MD - Estrogen		<1%							3%	3%	0.69%
Law - Narcotic			8%	9%		1%		4%		10%	3.39%
Clinic - Fertility											
Total	8.11%	6.11%	10.58%	9.70%		8.71%		10.24%	5.87%	11.12%	

How did you determine your prices?

Pricing Strategies

Consider: perfect competition vs. concentrated vs. monopoly

- Cost-based pricing
 - Value-based pricing
 - Competitive pricing
 - Penetration pricing
 - Skimming pricing
 - Psychological pricing
 - Dynamic pricing
 - Freemium pricing
 - Bundling
 - Price discrimination
-
- Perfect competition – no pricing strategy, price takers
 - Concentrated industry – competitive pricing, penetration pricing, strategic pricing
 - Monopoly – skimming, price discrimination

Strategic Pricing

		Firm B	
Firm A		High	Low
	High	(8,8)	(4,10)
	Low	(10,4)	(5,5)

Price Cooperation

Tacit Collusion

- When are firms most incentivized to tacitly collude?
- When can firms actually come to an unspoken agreement?
- When can it be sustained? Why would it break down?

Factors Making Sustaining Pricing Cooperation More Likely

- Incentives/Motivation: Gains from cooperation vs. competition
 - Inelastic industry demand (higher markups)
 - High cross-price elasticities (little differentiation)
 - Low entry threat
 - Fringe firms can't expand
 - High barriers to entry
- Achievability: Communication and monitoring ability
 - Homogeneous products
 - Concentrated market
 - Long history of interacting with each other
 - Demand and costs conditions are stable
 - Established norms
 - No secret pricing deals
- Sustainability: Threat of punishment and value of future dollars
 - Firms value the future a lot relative to the present
 - Little threat of bankruptcy or exit
 - Demand growing
 - Firms can adjust prices quickly
 - Multimarket contact (large scope for scale of response)
 - Patience, rationality

Price Competition

- Nothing wipes out profit like price competition.
- In most industries, companies choose prices, but the individual incentive to cut prices leads to prices that leave each participant worse off.
- Since it is not really a 1-shot game, it is possible to react to a competitor's pricing decision.
- When considering price cuts, bear in mind the entire impact of the cut on prices. This includes the impact of any anticipated competitive response. If a company can anticipate price matching or a more aggressive punishment by a rival, a price cut may be unprofitable.

Source: The Dynamics of Price Competition by Garicano and Gertner

Industry Conditions that affect the Degree of Price Competition

1. The number of companies in an industry
 - Cooperative pricing becomes more difficult as the number of companies increases
 - Coordination more difficult and less likely to happen
 - May lead to reduced incentives to punish a price cut
 - Not easy to observe rivals' prices as the number of companies grows
2. Differences among the companies in the industry
 - Cost structure, capacity, existing market share, product quality, characteristics and range
 - Difference make coordination more difficult
 - A company may not respond because losing a small share is better than the loss of margin on cutting prices
 - Another company may prefer low prices even if there is a competitive response
3. The short run gain from cutting prices
 - If short run gain from cutting prices is small, so is the incentive to “cheat” on an implicit high price agreement
 - Short run gains are low when there is no excess capacity – can't make it up in volume
 - Short run gains are low when products are highly differentiated – customers less like to switch over
 - Short run gains are high when the opposite exists
4. Price transparency
 - If secret price cuts are possible, the incentive to cut prices is higher
 - Implicit coordination is harder when companies cannot observe their rivals' competitive actions
 - The more transparent the market, the lower the incentive to cut prices as they are easily matched

Source: The Dynamics of Price Competition by Garicano and Gertner

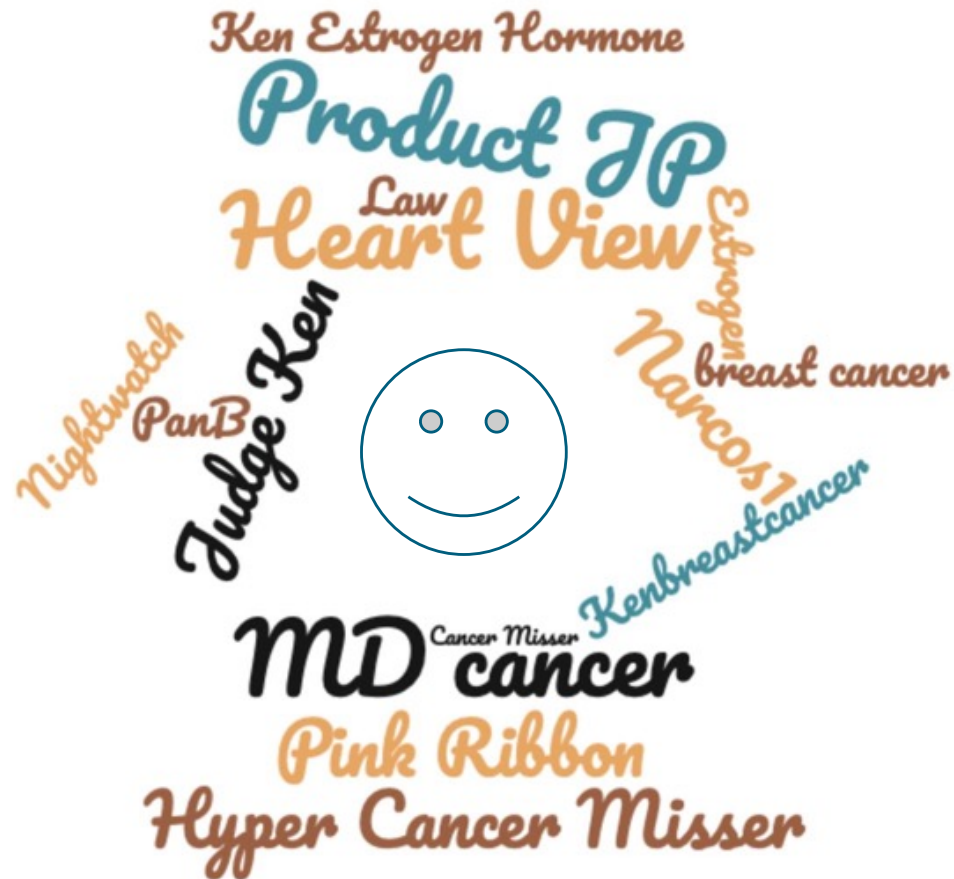
Changing the Rules of the Game

- Increase market transparency
 - Example: Create market segmentation
 - Example: Commitments to published price lists
 - Example: Advanced announcement of price changes
 - Example: Price leadership by a company that can play the price-setting role leaving all the other companies to imitate it
- Decrease the short run gains from price cuts,
 - Increase in product differentiation – consumers less likely to react to price cuts
 - Example: Frequent flier programs - if you are a BA frequent flier member, you are less likely to respond to a lower price by United Airlines on the same route, so United Airlines has less of an incentive to cut the prices.
 - Example: Most-favored-customer clauses

Source: The Dynamics of Price Competition by Garicano and Gertner

Product Designs

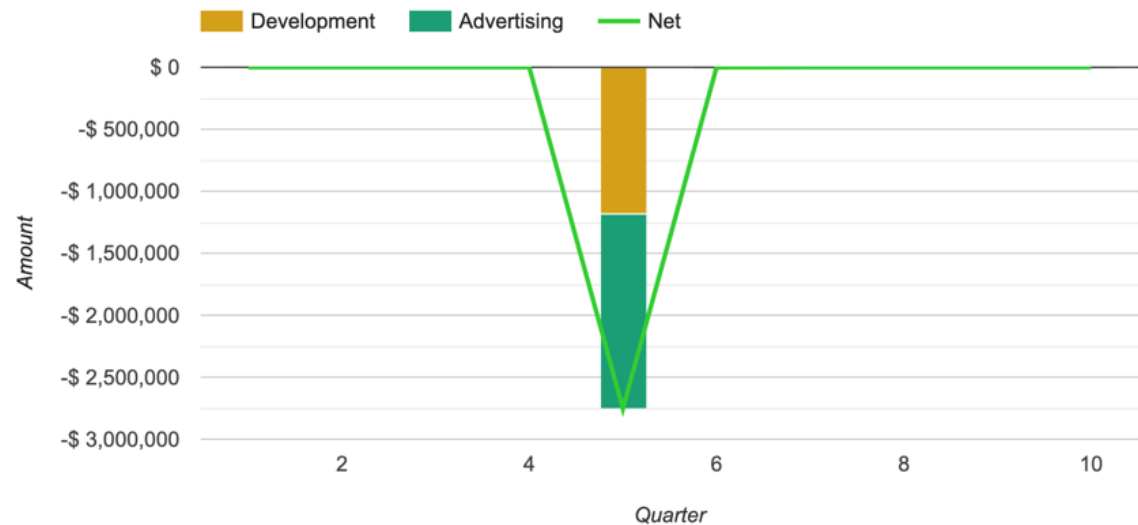
Be creative – name your products – only 1 chance to do so



Don't do these things

Retail Price: \$5,000
Units On Hand: 0

Heartbeat : Pulse only
Blood vessel motion : full profile
Toxicology : Barbiturate
Metabolic : Bilirubin
Cancer : Blood & Bone
Neurotoxins : Anatoxin-a
Motion : plus gait
Platform : wrist
GPS : GPS
Networking : 2.4 GHz
Power : Ni-Cd battery
Finish : Original
Material cost : \$995



Throughput : 9.04 units per day
Overhead per Unit : \$ 278.45

Status: In Development
Start date: 390.80
Ready date: -
Design days: 630

Retail sales:	\$0
Advertising:	- \$1,575,164
Development:	- \$1,180,000
Net:	\$-2,755,164

Participants:	20
Would buy:	100%
Best WTP:	\$44.89
Median WTP:	\$20.59

Don't do this either

Name : Pilot Factory in Dune

Owner : Dune

Location : Dune

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Capital expenditure : \$ 500,000

Daily expenditure rate : \$ 1

Operating income : \$ -3,082,881 

Inventory in process : 100 Units 
of Dune-MDBrstDune     
for Local DC in Dune 

Inventory in transit : 0 Units 
of 0 Products 
for 0 DCs 

Throughput : 0.01 units per day

Overhead per Unit : \$ 6,706.94

Suspended on day 910 

Product Analysis Report

Don't do this either

Product	Manufacturer	Distribution Center	Region	Owner	Price	On hand	Retail sales	♥	○	💧	△	H	✱	C	💬
Heart View	Dune		Local DC in Dune	Dune	Dune	\$750	0	6,920	♥	-	-	-	-	-	-
Heart View	Globex		Local DC in Globex	Globex	Globex	\$600	23	7,527	♥	-	-	-	-	-	-
Product28	Panem		Local DC in Globex	Globex	Globex	\$700	0	300	♥	-	-	-	LH	-	-
Panem Cancer	Panem		Local DC in Globex	Globex	Globex	\$650	0	100	♥	🎯	-	-	-	-	©
Fertility Estro	Gotham		Local DC in Globex	Globex	Globex	\$700	0	1,400	♥	-	-	-	H	-	-
Heart View	Gotham		Local DC in Gotham	Gotham	Gotham	\$720	24	8,676	♥	-	-	-	-	-	-
Fertility Estro	Gotham		Local DC in Gotham	Gotham	Gotham	\$700	25	1,075	♥	-	-	-	H	-	-
Heart View	Metropolis		Local DC in Metropolis	Metropolis	Metropolis	\$700	0	8,000	♥	-	-	-	-	-	-
Heart View	Panem		Local DC in Panem	Panem	Panem	\$720	306	17,394	♥	-	-	-	-	-	-
Product28	Panem		Local DC in Panem	Panem	Panem	\$500	0	1,300	♥	-	-	-	LH	-	-
Panem Cancer	Panem		Local DC in Panem	Panem	Panem	\$500	0	1,200	♥					-	©
Heart View	Serenity		Local DC in Serenity	Serenity	Serenity	\$700	0	8,000	♥	-	-	-	-	-	-
Heart View	Vulcan		Local DC in Vulcan	Vulcan	Vulcan	\$718	177	8,423	♥	-	-	-	-	-	-
MD Fertility	Vulcan		Local DC in Vulcan	Vulcan	Vulcan	\$774	107	593	♥	-	-	-	H	-	-
Heart View	Westeros		Local DC in Westeros	Westeros	Westeros	\$410	193	7,707	♥	-	-	-	-	-	-
Heart View	Zion		Local DC in Zion	Zion	Zion	\$600	0	8,700	♥	-	-	-	-	-	-
MDferLH	Dune		Local DC in Dune	Dune	Dune	\$780	0	400	-	-	-	-	H	-	-
MDBrstDune	Dune		Local DC in Dune	Dune	Dune	\$1,200	0	300	-	-	-	-	-	-	©
LawDune	Dune		Local DC in Dune	Dune	Dune	\$1,000	0	300	-	-	-	△	-	-	-
Fertility Type LH FSH	Zion		Local DC in Dune	Dune	Dune	\$350	100	0	-	-	-	-	H+	-	-
GlobexNarcotics	Globex		Local DC in Globex	Globex	Globex	\$850	0	1,300	-	-	-	△	-	-	-
GlobexCancer	Globex		Local DC in Globex	Globex	Globex	\$1,000	0	1,250	-	-	-	-	-	-	©

The business is a going concern

Don't shut down at the end of the game!!

Pilot Factory in Vulcan [Vulcan] [PROCESSING] ✕

Shutdown is pending completion of current batch.

Name : Pilot Factory in Vulcan

Owner : Vulcan

Location : Vulcan

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Capital expenditure : \$ 100,000

Daily expenditure rate : \$ 2,000

Operating income : \$ -2,500,240

Inventory in process : 100 Units
of Vulcan-BC Beater
for Local DC in Vulcan

Inventory in transit : 0 Units
of 0 Products
for 0 DCs

Throughput : 7.49 units per day

Overhead per Unit : \$ 269.56

Suspended on day 910

2023 Vulcan
Not our Vulcan!

11 Incoming Shipping Agreements

10 records Search:

Id	Origin	Product	Reorder point	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
118		BC Beater	10	20 @ \$100	0	1	\$ 930	ACTIVE
4		Heart View	0	100 @ \$100	-	64	\$ 450	TERMINATED
29		BC Beater	20	100 @ \$315	-	35	\$ 715	TERMINATED
56		Pablo	5	5 @ \$900	-	1	\$ 130	TERMINATED
65		Narco Westeros	1	5 @ \$1,150	-	1	\$ -200	TERMINATED
66		Product17	0	5 @ \$135	-	2	\$ -105	TERMINATED
83		Lymph2Lymph	0	10 @ \$900	-	7	\$ -30	TERMINATED
89		Narco Westeros	0	100 @ \$700	-	0	\$ 250	TERMINATED
90		Narco Westeros	0	5 @ \$700	-	3	\$ 250	TERMINATED
96		Heart View	40	40 @ \$600	-	0	\$ -10	TERMINATED

8 Inventories

10 records

Product	\$	Units
Name	Price	On hand
BC Beater	1,300	0
Heart View	750	0
Heart View	700	0
hot sauce	850	0
Lymph2Lymph	1,100	0
Narco Westeros	1,200	0
Pablo	1,300	0
Product17	50	10

Don't do this

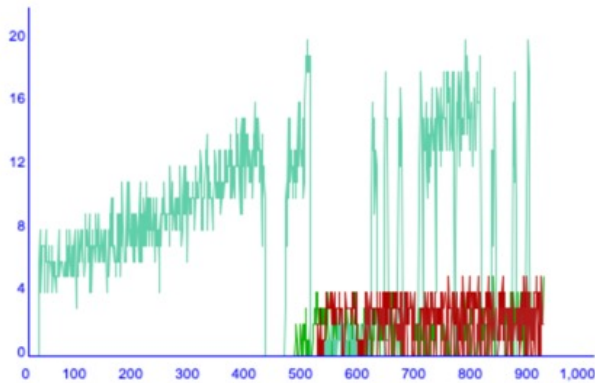
Your company is a going concern.

	\$	Quarter 12
Sales revenue	\$	0
Cost of goods sold	\$	0
Gross profit	\$	0
Advertising expense	\$	0
Selling expense	\$	0
Distribution center operating expense	\$	182,000
Total selling expense	\$	182,000
Research and development expense	\$	0
Distribution center depreciation expense	\$	41,667
Idle factory expense	\$	458,333
Other operating expenses	\$	500,000
Operating income	\$	(682,000)
Interest revenue	\$	204
Interest expense on emergency loan	\$	16,061
Net interest and fees revenue (expense)	\$	(15,857)
Total net income before tax	\$	(697,857)
Income tax expense	\$	(244,249)
Net income after tax and before extraordinary items	\$	(453,607)
Net income	\$	(453,607)

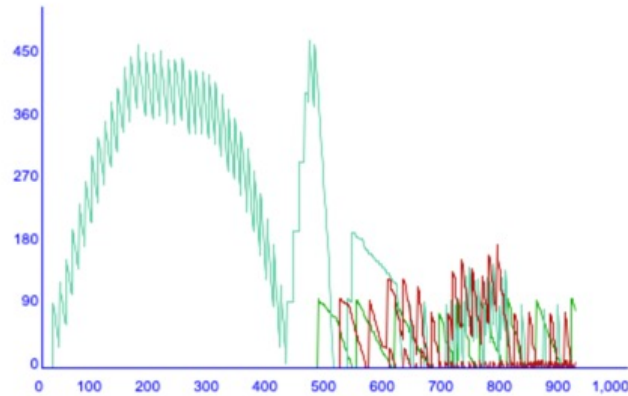
Managing Supply and Demand

Stockouts and Idle Time

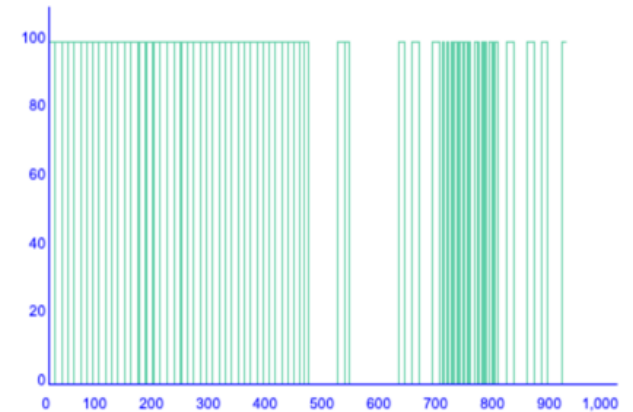
Retail Sales



On Hand Inventory



Work In Process Inventory

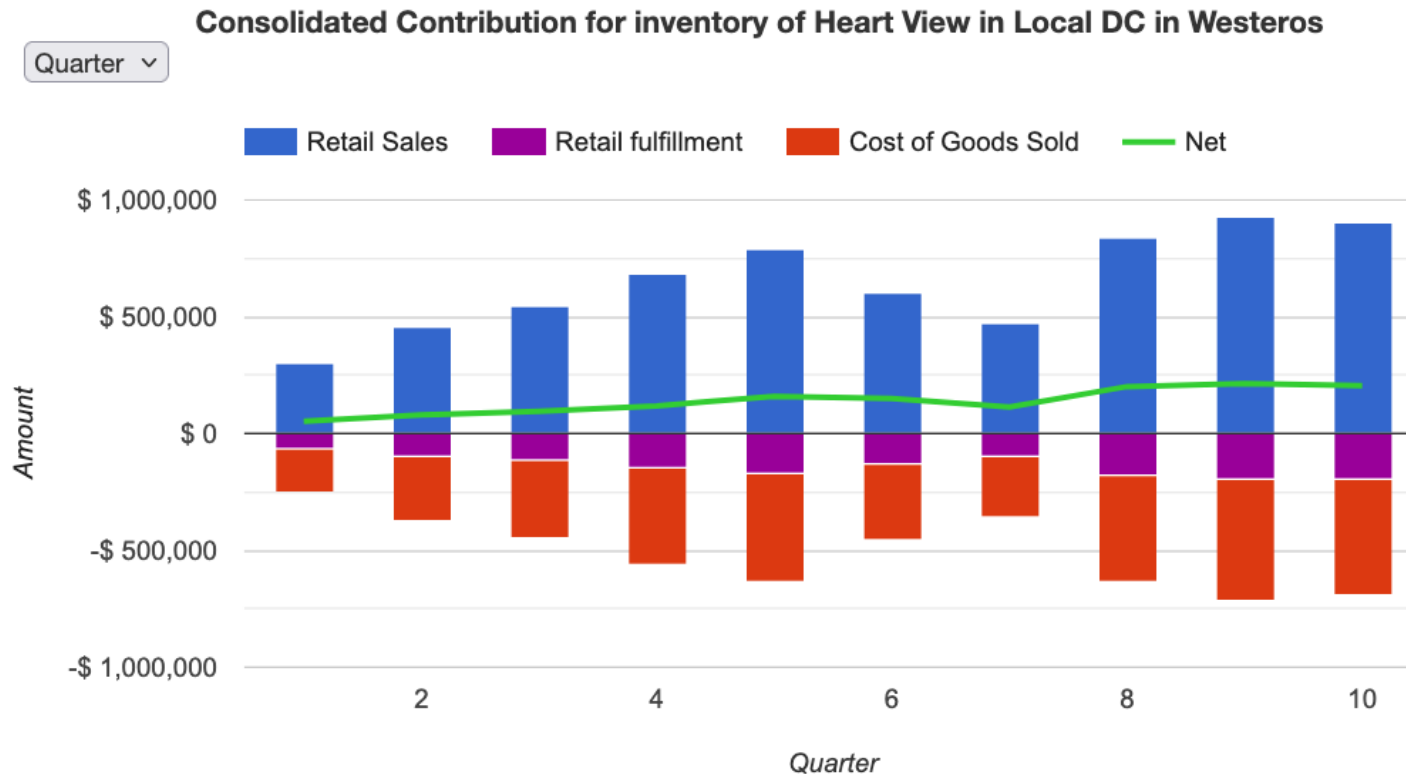


Top Products by Contribution Margin

World	Team	Product	Contribution ▼	Retail units	Retail revenue	Average retail price	Current retail price	Average service rate	Days SA active	Current ROP	Cumulative advertising	Advertising expense
1	Gotham	     	\$1,670,756	11,115	\$7,780,500	\$700	\$700	95.33%	910	500	\$9,507	\$9,507
1	Panem	     	\$1,467,921	9,989	\$6,992,300	\$700	\$700	89.30%	910	150	\$0	\$0
1	B612	     	\$1,306,922	9,817	\$6,822,400	\$695	\$700	86.16%	910	700	\$25,603	\$25,603
1	Globex	     	\$1,273,782	11,092	\$7,765,320	\$700	\$700	96.70%	910	0	\$6,283	\$6,283
1	Zion	     	\$1,197,456	11,600	\$8,120,000	\$700	\$700	99.01%	910	200	\$0	\$0
1	Vulcan	     	\$1,121,964	9,987	\$6,993,650	\$700	\$700	93.93%	910	200	\$0	\$0
1	Dune	     	\$816,208	6,068	\$4,243,410	\$699	\$700	65.76%	910	0	\$0	\$0
1	Westeros	     	\$601,046	5,200	\$3,525,000	\$678	\$750	78.19%	910	50	\$0	\$0

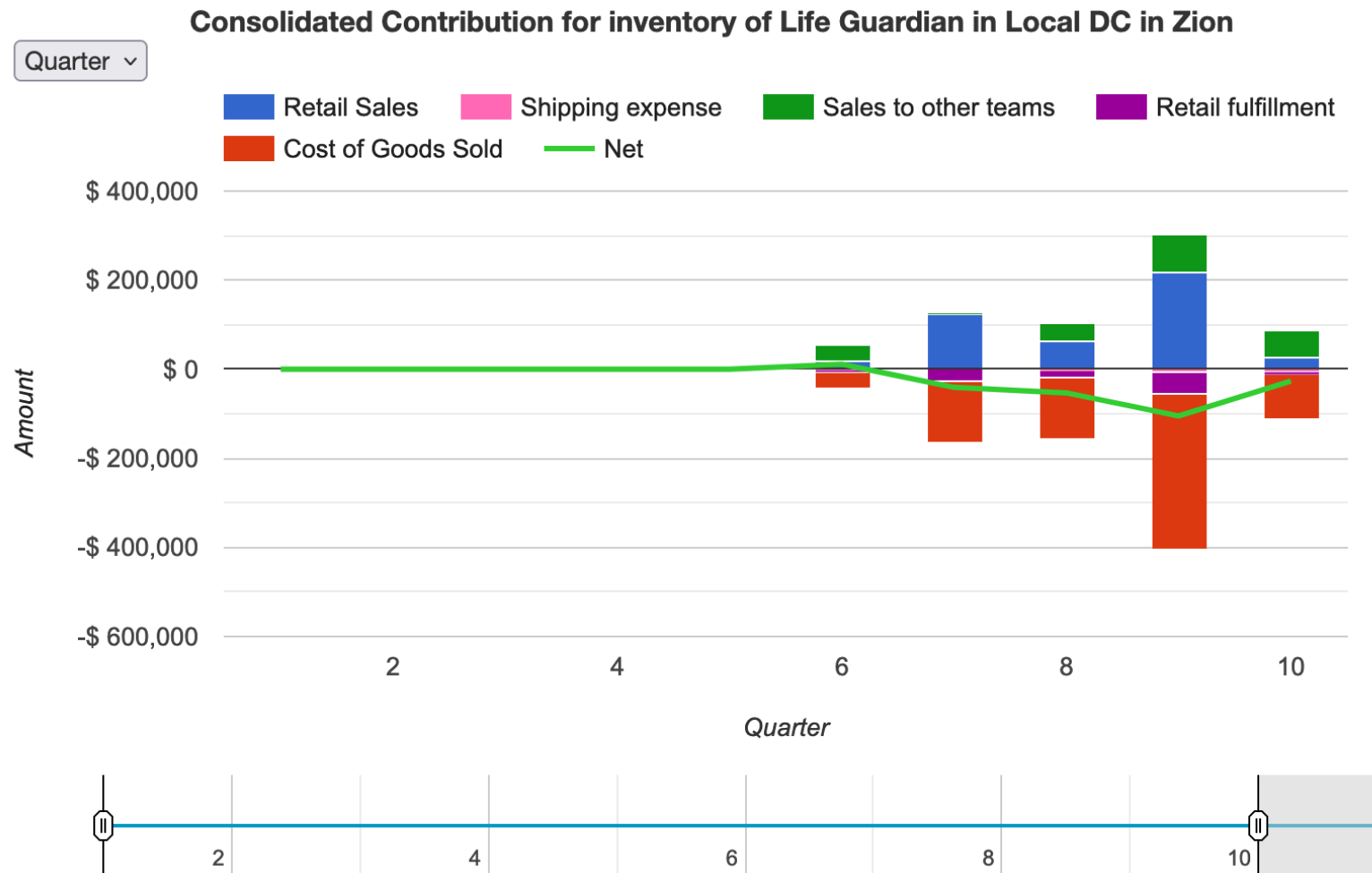
Product Contribution

Westeros Heart View (Last Year)



Product Contribution

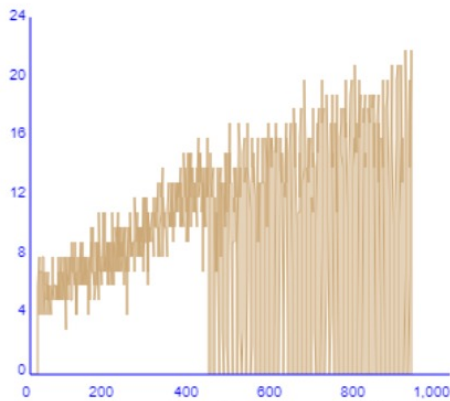
Zion Life Guardian (last year)



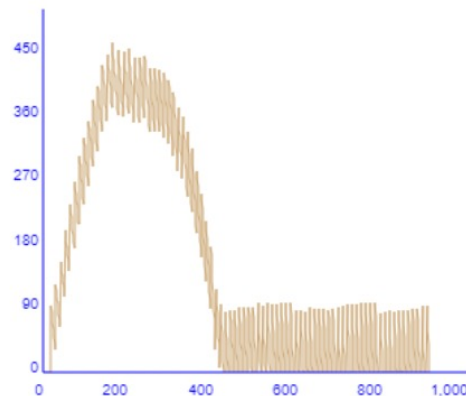
Do Nothing Team: Supply and Demand

Sold through built up inventory and factory couldn't keep up

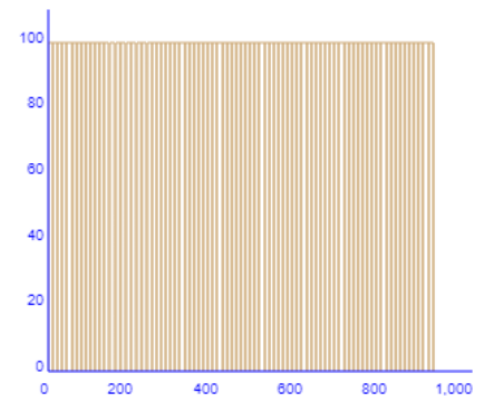
Retail Sales



On Hand Inventory



Work In Process Inventory



Do Nothing Team: Finances

Stagnated at steady negative state, no R&D for new products, no capital investment, no growth

	\$	Quarter 10	Quarter 9	Quarter 8	Quarter 7	Quarter 6	Quarter 5
Sales revenue	\$	581,700	593,600	574,700	565,600	582,400	716,100
Cost of goods sold	\$	356,045	363,329	351,761	346,191	356,474	438,309
Gross profit	\$	225,655	230,271	222,939	219,409	225,926	277,791
Selling expense	\$	124,650	127,200	123,150	121,200	124,800	153,450
Distribution center operating expense	\$	182,000	182,000	182,000	182,000	182,000	182,000
Total selling expense	\$	306,650	309,200	305,150	303,200	306,800	335,450
Distribution center depreciation expense	\$	41,667	41,667	41,667	41,667	41,667	41,667
Idle factory expense	\$	0	0	0	0	0	0
Other operating expenses	\$	41,667	41,667	41,667	41,667	41,667	41,667
Operating income	\$	(122,662)	(120,596)	(123,878)	(125,458)	(122,540)	(99,325)
Interest revenue	\$	10,229	10,745	11,252	11,757	12,261	12,138
Net interest and fees revenue (expense)	\$	10,229	10,745	11,252	11,757	12,261	12,138
Total net income before tax	\$	(112,433)	(109,851)	(112,626)	(113,700)	(110,279)	(87,187)
Income tax expense	\$	(39,351)	(38,447)	(39,419)	(39,795)	(38,597)	(30,515)
Net income after tax and before extraordinary items	\$	(73,082)	(71,403)	(73,207)	(73,906)	(71,682)	(56,672)
Net income	\$	(73,082)	(71,403)	(73,207)	(73,906)	(71,682)	(56,672)

	%	Quarter 10	Quarter 9	Quarter 8	Quarter 7	Quarter 6	Quarter 5
Sales revenue	%	100	100	100	100	100	100
Cost of goods sold	%	61.21	61.21	61.21	61.21	61.21	61.21
Gross profit	%	38.79	38.79	38.79	38.79	38.79	38.79
Selling expense	%	21.43	21.43	21.43	21.43	21.43	21.43
Distribution center operating expense	%	31.29	30.66	31.67	32.18	31.25	25.42
Total selling expense	%	52.72	52.09	53.10	53.61	52.68	46.84
Distribution center depreciation expense	%	7.16	7.02	7.25	7.37	7.15	5.82
Idle factory expense	%	0	0	0	0	0	0
Other operating expenses	%	7.16	7.02	7.25	7.37	7.15	5.82
Operating income	%	-21.09	-20.32	-21.56	-22.18	-21.04	-13.87
Interest revenue	%	1.76	1.81	1.96	2.08	2.11	1.69
Net interest and fees revenue (expense)	%	1.76	1.81	1.96	2.08	2.11	1.69
Total net income before tax	%	-19.33	-18.51	-19.60	-20.10	-18.94	-12.18
Income tax expense	%	-6.76	-6.48	-6.86	-7.04	-6.63	-4.26
Net income after tax and before extraordinary items	%	-12.56	-12.03	-12.74	-13.07	-12.31	-7.91
Net income	%	-12.56	-12.03	-12.74	-13.07	-12.31	-7.91

	\$	Quarter 10	Quarter 9	Quarter 8	Quarter 7	Quarter 6	Quarter 5
Assets							
Current assets:							
Cash	\$	1,322,362	1,408,533	1,474,688	1,524,086	1,596,379	1,676,618
Accounts receivable	\$	210,000	188,300	167,300	192,500	204,400	184,100
Work in process inventory	\$	15,886	37,328	30,925	24,522	18,119	11,715
Finished goods inventory	\$	42,845	13,282	33,848	42,845	46,273	59,983
Tax benefit	\$	410,490	371,138	332,691	293,272	253,478	214,880
Total current assets	\$	2,001,584	2,018,582	2,039,452	2,077,226	2,118,648	2,147,297
Noncurrent assets:							
Land	\$	200,000	200,000	200,000	200,000	200,000	200,000
Property plant and equipment - factories:							
Property plant and equipment - factories	\$	100,000	100,000	100,000	100,000	100,000	100,000
Accumulated depreciation - factories	\$	(16,667)	(15,000)	(13,333)	(11,667)	(10,000)	(8,333)
Net Property plant and equipment - factories	\$	83,333	85,000	86,667	88,333	90,000	91,667
Property plant and equipment - distribution centers:							
Property plant and equipment - distribution centers	\$	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000	2,500,000
Accumulated depreciation - distribution centers	\$	(416,667)	(375,000)	(333,333)	(291,667)	(250,000)	(208,333)
Net Property plant and equipment - distribution centers	\$	2,083,333	2,125,000	2,166,667	2,208,333	2,250,000	2,291,667
Total assets	\$	4,368,250	4,428,582	4,492,785	4,573,893	4,658,648	4,730,630
Liabilities and shareholders' equity							
Liabilities:							
Current liabilities:							
Raw materials payable	\$	30,000	20,000	20,000	30,000	30,000	30,000
Other payables	\$	100,600	97,850	90,650	88,550	99,400	99,700
Total current liabilities	\$	130,600	117,850	110,650	118,550	129,400	129,700
Noncurrent liabilities:							
Noncurrent liabilities	\$	0	0	0	0	0	0
Shareholder equity:							
Contributed capital	\$	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000	5,000,000
Retained earnings	\$	(762,350)	(689,268)	(617,865)	(544,657)	(470,752)	(399,070)
Total shareholder equity	\$	4,237,650	4,310,732	4,382,135	4,455,343	4,529,248	4,600,930
Total liabilities and shareholders' equity	\$	4,368,250	4,428,582	4,492,785	4,573,893	4,658,648	4,730,630

Production Game Wish List

- What did you wish you could do?

Production and Distribution

Bench, Line, Cell

DCs

Production and Distribution

- What type of factory do you use?
- How much capital?
- How much labor?
- What batch size?
- How do you distribute to retail customers?
- How do you distribute to your own DCs?
- How do you distribute to wholesale customer DCs?
- Do you use mail or container?

Production Technologies

Technology	A	α	β	Technology Description
Benches	0.009	0.10	0.85	Lab benches with general-purpose equipment. Each table is manned by a skilled worker who produces an entire product. Setup time for new batches = 0.05 days
Production Line	0.010	0.30	0.75	A row of tables with specialized equipment and each worker assigned a specific portion of the production sequence. Workers are relatively unskilled. Setup time for new batches = 0.50 days
Automated Cell	0.020	0.80	0.30	A robot that feeds material to automated production machines clustered around it. There is support staff but no workers performing actual production. Setup time for new batches = 1 day

Minimum Capital and Labor Requirements

Factory Type	Minimum Capital Investment	Minimum Daily Labor Expenditure
Bench	\$1	\$1 (\$0 to shut down)
Production Line	\$500,000	
Automated Cell	\$3,000,000	

NEVER put \$1 or anything close to \$1 for labor – you will ruin your factory.

Production Model

Daily Throughput (100 unit batches in this case)

		Bench						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		2.310	4.161	9.044	10.552	13.455	24.123	43.065
200,000		2.476	4.459	9.690	11.305	14.414	25.832	46.085
300,000		2.578	4.643	10.089	11.770	15.006	26.887	47.946
400,000		2.654	4.778	10.382	12.112	15.441	27.661	49.311
500,000		2.713	4.886	10.615	12.383	15.786	28.276	50.396
600,000		2.763	4.975	10.809	12.610	16.075	28.789	51.299
700,000		2.806	5.052	10.976	12.804	16.322	29.229	52.075
800,000		2.844	5.120	11.123	12.975	16.540	29.616	52.757
900,000		2.877	5.181	11.254	13.128	16.734	29.962	53.365
1,000,000		2.908	5.235	11.372	13.266	16.910	30.275	53.915
2,000,000		3.116	5.610	12.184	14.211	18.113	32.412	57.673
3,000,000		3.245	5.841	12.685	14.795	18.855	33.731	59.988
4,000,000		3.340	6.011	13.052	15.224	19.400	34.698	61.685
5,000,000		3.415	6.147	13.345	15.564	19.833	35.467	63.033

		Line						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		11.682	18.894	34.357	38.424	45.569	66.333	90.984
600,000		12.298	19.851	35.941	40.151	47.522	68.780	93.703
700,000		12.843	20.693	37.325	41.655	49.218	70.881	96.009
800,000		13.333	21.448	38.556	42.992	50.719	72.725	98.010
900,000		13.779	22.134	39.668	44.196	52.068	74.368	99.776
1,000,000		14.190	22.764	40.683	45.294	53.295	75.850	101.357
2,000,000		17.188	27.307	47.837	52.990	61.807	85.856	111.700
3,000,000		19.198	30.304	52.404	57.861	67.118	91.861	117.649
4,000,000		20.749	32.591	55.809	61.473	71.020	96.160	121.796
5,000,000		22.028	34.458	58.542	64.360	74.115	99.505	124.960

		Cell						
		Daily Throughput in units, including setup time						
K ↓ vs L_daily →		500	1,000	2,500	3,000	4,000	8,000	16,000
100,000		na	na	na	na	na	na	na
200,000		na	na	na	na	na	na	na
300,000		na	na	na	na	na	na	na
400,000		na	na	na	na	na	na	na
500,000		na	na	na	na	na	na	na
600,000		na	na	na	na	na	na	na
700,000		na	na	na	na	na	na	na
800,000		na	na	na	na	na	na	na
900,000		na	na	na	na	na	na	na
1,000,000		na	na	na	na	na	na	na
2,000,000		na	na	na	na	na	na	na
3,000,000		75.960	79.550	83.662	84.396	85.499	87.892	89.937
4,000,000		79.909	83.041	86.570	87.193	88.126	90.136	91.837
5,000,000		82.623	85.409	88.513	89.058	89.871	91.613	93.079

Production Model

Manufacturing Overhead per Unit

Bench							
Manufacturing Overhead per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	224.33	244.74	278.45	286.04	298.64	332.39	371.96
200,000	216.72	232.49	261.78	268.61	280.05	311.11	347.98
300,000	215.23	227.22	253.24	259.55	270.22	299.59	334.85
400,000	216.03	224.63	247.86	253.75	263.80	291.87	325.96
500,000	218.02	223.43	244.15	249.66	259.18	286.16	319.31
600,000	220.72	223.08	241.45	246.63	255.68	281.70	314.04
700,000	223.87	223.31	239.45	244.31	252.92	278.08	309.71
800,000	227.35	223.93	237.94	242.50	250.70	275.07	306.06
900,000	231.05	224.85	236.79	241.08	248.88	272.51	302.91
1,000,000	234.93	226.00	235.94	239.95	247.38	270.30	300.16
2,000,000	278.00	243.55	235.26	236.87	241.06	258.12	283.78
3,000,000	323.41	265.25	240.41	239.91	241.29	253.46	275.88
4,000,000	369.10	288.22	247.66	245.18	243.95	251.67	271.26
5,000,000	414.60	311.67	255.96	251.58	247.85	251.38	268.36

Line							
Manufacturing Overhead per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	50.64	57.77	75.43	80.46	89.79	121.98	176.86
600,000	49.59	55.91	72.62	77.46	86.48	117.91	171.93
700,000	48.92	54.52	70.41	75.10	83.88	114.67	167.99
800,000	48.49	53.46	68.64	73.19	81.75	112.02	164.74
900,000	48.25	52.63	67.18	71.61	79.99	109.79	162.01
1,000,000	48.14	51.97	65.95	70.28	78.49	107.89	159.67
2,000,000	50.40	50.03	59.92	63.53	70.64	97.45	146.52
3,000,000	54.66	51.13	58.19	61.34	67.78	93.07	140.67
4,000,000	59.40	53.16	57.92	60.72	66.64	90.81	137.38
5,000,000	64.27	55.60	58.35	60.84	66.33	89.60	135.37

Cell							
Manufacturing Overhead per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	na	na	na	na	na	na	na
600,000	na	na	na	na	na	na	na
700,000	na	na	na	na	na	na	na
800,000	na	na	na	na	na	na	na
900,000	na	na	na	na	na	na	na
1,000,000	na	na	na	na	na	na	na
2,000,000	na	na	na	na	na	na	na
3,000,000	13.82	19.48	36.45	42.06	53.21	97.27	184.01
4,000,000	15.43	20.86	37.34	42.81	53.70	96.88	182.20
5,000,000	17.14	22.43	38.59	43.97	54.70	97.32	181.74

Production Model

Manufacturing Overhead + Materials Cost per Unit (\$100 in this case)

Bench							
Manufacturing Overhead plus Materials Cost per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	324.33	344.74	378.45	386.04	398.64	432.39	471.96
200,000	316.72	332.49	361.78	368.61	380.05	411.11	447.98
300,000	315.23	327.22	353.24	359.55	370.22	399.59	434.85
400,000	316.03	324.63	347.86	353.75	363.80	391.87	425.96
500,000	318.02	323.43	344.15	349.66	359.18	386.16	419.31
600,000	320.72	323.08	341.45	346.63	355.68	381.70	414.04
700,000	323.87	323.31	339.45	344.31	352.92	378.08	409.71
800,000	327.35	323.93	337.94	342.50	350.70	375.07	406.06
900,000	331.05	324.85	336.79	341.08	348.88	372.51	402.91
1,000,000	334.93	326.00	335.94	339.95	347.38	370.30	400.16
2,000,000	378.00	343.55	335.26	336.87	341.06	358.12	383.78
3,000,000	423.41	365.25	340.41	339.91	341.29	353.46	375.88
4,000,000	469.10	388.22	347.66	345.18	343.95	351.67	371.26
5,000,000	514.60	411.67	355.96	351.58	347.85	351.38	368.36

Line							
Manufacturing Overhead plus Materials Cost per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	150.64	157.77	175.43	180.46	189.79	221.98	276.86
600,000	149.59	155.91	172.62	177.46	186.48	217.91	271.93
700,000	148.92	154.52	170.41	175.10	183.88	214.67	267.99
800,000	148.49	153.46	168.64	173.19	181.75	212.02	264.74
900,000	148.25	152.63	167.18	171.61	179.99	209.79	262.01
1,000,000	148.14	151.97	165.95	170.28	178.49	207.89	259.67
2,000,000	150.40	150.03	159.92	163.53	170.64	197.45	246.52
3,000,000	154.66	151.13	158.19	161.34	167.78	193.07	240.67
4,000,000	159.40	153.16	157.92	160.72	166.64	190.81	237.38
5,000,000	164.27	155.60	158.35	160.84	166.33	189.60	235.37

Cell							
Manufacturing Overhead plus Materials Cost per unit							
K ↓ vs L_daily →	500	1,000	2,500	3,000	4,000	8,000	16,000
100,000	na	na	na	na	na	na	na
200,000	na	na	na	na	na	na	na
300,000	na	na	na	na	na	na	na
400,000	na	na	na	na	na	na	na
500,000	na	na	na	na	na	na	na
600,000	na	na	na	na	na	na	na
700,000	na	na	na	na	na	na	na
800,000	na	na	na	na	na	na	na
900,000	na	na	na	na	na	na	na
1,000,000	na	na	na	na	na	na	na
2,000,000	na	na	na	na	na	na	na
3,000,000	113.82	119.48	136.45	142.06	153.21	197.27	284.01
4,000,000	115.43	120.86	137.34	142.81	153.70	196.88	282.20
5,000,000	117.14	122.43	138.59	143.97	154.70	197.32	281.74

Build New Factory

Globex Region

As of day 368.31

1 Inventory

10 records Search:

Product	DC	\$	Units	Last date
Name		Price	On hand	Any
Heart View	Globex	700	207	367.94

Showing 1 to 1 of 1 entries

1 Factory

10 records Search:

ADD NEW

Team	Name	Region	Status	WIP Units	Tech	Cap EX (\$)	Daily EX (\$)
				Total		Total	Total
Globex	Pilot Factory in Globex	Globex	PROCESSING	100		100,000	2,500

Showing 1 to 1 of 1 entries

1 Distribution Center

10 records Search:

ADD NEW

Team	Name	Region	Status
			Current
Globex	Local DC in Globex	Globex	READY

Showing 1 to 1 of 1 entries

Start New Factory

Name

Globex

Factory 1

Land Cost

\$

100,000.00

Capital Expenditure

\$

Technology

===Select===

Build Duration

90

days

Daily Expenditure

\$

SA Clone Factory List

None

CANCEL

CREATE

Confirm Transaction

Factory Transaction Detail

Factory Name : Globex-New Factory

Land Cost : \$100,000.00

Capital Exp : \$10,000.00

Technology : Bench

Build Duration : 90

Daily Exp : \$0.00

When :

Now

Confirm

Cancel

Increase Capital

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : \$ 2,500

Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703

Inventory in process : 100 Units
of Globex-Heart View
for Local DC in Globex

Inventory in transit : 100 Units
of Globex-Heart View
for Local DC in Globex

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

As of day 366.14

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Capital expenditure : \$ 100,000

\$ ✓ ✗

Confirm Transaction

Pending Capital Expenditure will be effective on day
411.00

Factory Name : Pilot Factory in Arryn
Current Cap Exp : 100,000
Requested Cap Exp : 1,000.00
Effective Day : 411.00
When : Now

Confirm Cancel

Change Labor

Pilot Factory in Globex [Globex] [PROCESSING]

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : **\$ 2,500**

Capital expenditure : \$ 100,000

Operating income : \$ -1,081,703

Inventory in process : 100 Units

of Globex-Heart View

for Local DC in Globex

Inventory in transit : 100 Units

of Globex-Heart View

for Local DC in Globex

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

As of day 366.14

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	33	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Daily expenditure : \$ 2,500

\$ 0

Confirm Transaction

Factory Change Daily Exp Transaction

Factory Name : Pilot Factory in Arrakis

Current Daily Exp : 2,500.00

Requested Daily Exp : 2,000.00

When : Now

Confirm

Cancel

Shut Down Factory

Pilot Factory in Globex [Globex] [PROCESSING]

Shutdown is pending completion of current batch.

As of day 439.19

Name : Pilot Factory in Globex

Owner : Globex

Location : Globex

Status : PROCESSING

Technology : Bench

Land expenditure : \$ 100,000

Daily expenditure : \$ 2,500

Capital expenditure : \$ 100,000

Operating income : \$ -1,295,540

Inventory in process : 100 Units
of Globex-Heart View
for Local DC in Globex

Inventory in transit : 100 Units
of Globex-Heart View
for Local DC in Globex

Throughput : 9.04 units per day

Overhead per Unit : \$ 278.45

1 Outgoing Shipping Agreement

10 records

Search:

Id	Destination	Product	Priority	Terms	Days delayed	Batches	Projected contribution per unit	Status
		Name					Margin	
1		Heart View	0	100 @ \$100	0	39	\$ -50	ACTIVE

Showing 1 to 1 of 1 entries

Close

Build New Distribution Center

Globex Region

As of day 368.31

1 Inventory

10 records

Search:

Product	DC	\$	Units	Last date																
Name		Price	On hand	Any																
Heart View		700	207	367.94																

Showing 1 to 1 of 1 entries

1 Factory

10 records

Search:

	Team	Name	Region	Status	WIP Units	Tech	Cap EX (\$)	Daily EX (\$)
							Total	Total
	Globex	Pilot Factory in Globex	Globex	PROCESSING	100		100,000	2,500

Showing 1 to 1 of 1 entries

1 Distribution Center

10 records

Search:

	Team	Name	Region	Status
				Current
	Globex	Local DC in Globex	Globex	READY

Showing 1 to 1 of 1 entries

Start New Distribution Center

Name

Globex- DC 1

Region

Globex

Land Cost

\$100,000

Capital Expenditure

\$2,500,000

Build Duration

60 days

Daily Expenditure

\$2,000

Fulfillment Cost

\$10 + 20 % of retail price

SA Clone Building List

None

CANCEL

CREATE

Confirm Transaction

DC Transaction Detail

DC Name : Globex-DC 1

Land Cost : \$100,000

Capital Expenditure : \$2,500,000

Build Duration : 60 days

Daily Expenditure : \$2,000

When : Now

Confirm

Cancel

Shut Down Distribution Center

Local DC in Globex [Globex] [READY]

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎
Operating income : \$ 2,564,186 📊

Total advertising : \$ 44,062.84 📊
Total inventory : 50 units 📊
On-hand inventory : 0 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,900 units 📊

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎

Total advertising : \$ 44,062.84 📊
Total inventory : 50 units 📊
On-hand inventory : 0 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,900 units 📊

Pending status : SHUTDOWN ✔ ✖

Operating income : \$ 2,564,186 📊

Confirm Transaction



Stop accepting orders now and shut down DC once
there is no remaining inventory?

When :

Confirm

Cancel

Confirm Transaction



Distribution Center ShutDown is
Scheduled Successfully.

Close

Local DC in Globex [Globex] [READY]



Shutdown is pending. Not placing new orders.

Name : Local DC in Globex
Owner : Globex
Location : Globex
Status : READY ✎
Operating income : \$ 2,558,773 📊

Total advertising : \$ 44,081.55 📊
Total inventory : 75 units 📊
On-hand inventory : 25 units 📊
Price range : \$ 600 to \$ 1,000 📊
Total retail sales : 11,925 units 📊

As of day 919.87 🔄

Distribution

- Direct from factory to DC
- Factory to central DC
 - Hub and Spoke
- Priorities

Marketing

Bass Model, Imitators, Innovators, Advertising

Marketing

All regions have the same size markets, with the following exceptions:

- Metropolis has larger markets – up to 2x other regions' markets.
- Serenity is the only region with military markets.
- Serenity has very small non-military markets.

Customer arrival rates determined by the Bass model and advertising spend

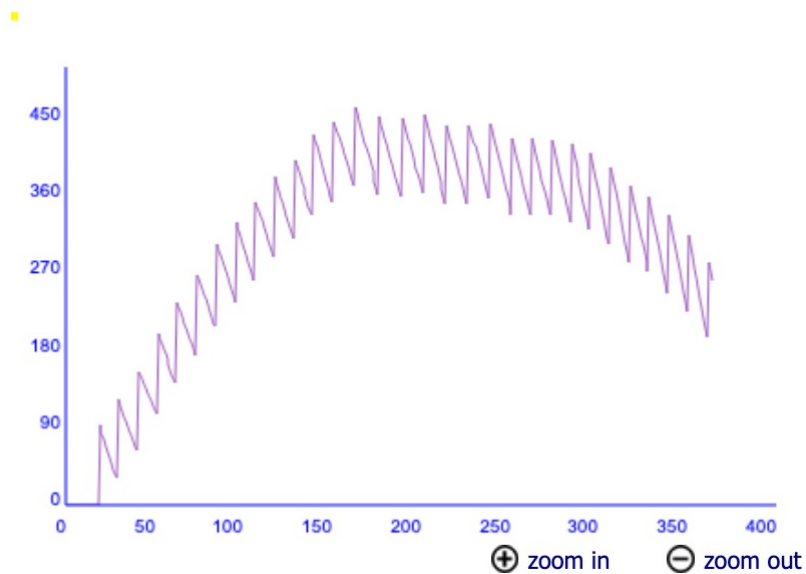
- Innovators (p parameter)
- Imitators (q parameter)
- Advertising (\$500 per additional p)

Distribution

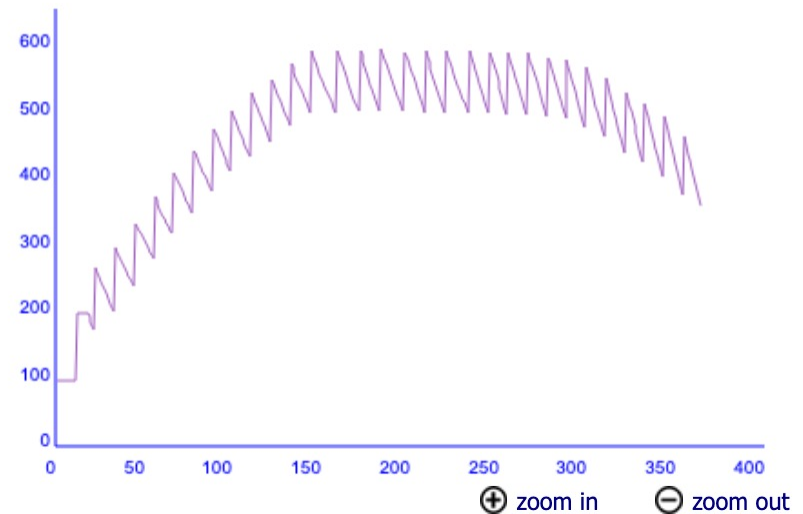
- Factory to DC
- DC to DC
- Retail customers only buy from DCs

Inventory Balances in Year 1

End-of-day On-hand Inventory at Local DC in Westeros



End-of-day Total Inventory at Local DC in Westeros



Daily Sales in Year 1

Name : Local DC in Westeros

Owner : Westeros

Location : Westeros

Status : READY

Operating income : \$ 381,983 

Total advertising : \$ 0 

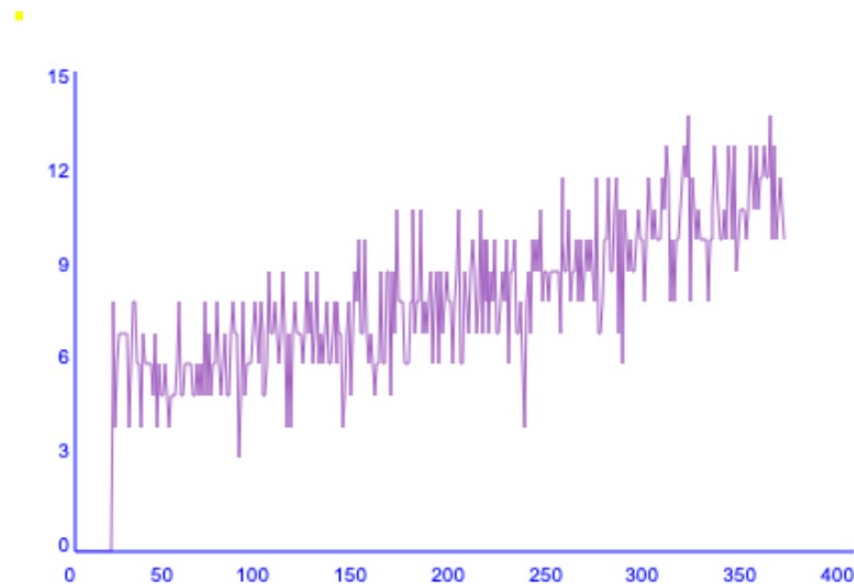
Total inventory : 363 units 

On-hand inventory : 263 units 

Price : \$ 700 

Total retail sales : 2,837 units 

Daily Retail Sales from Local DC in Westeros



Estimating Demand – Bass Model

Who is coming to market and when will they buy?

- Innovators
 - Price shop (in the region)
 - Purchase based on price
 - Purchase the product that gives them the greatest WTP - Price
 - Daily Arrivals = $p * \text{remaining market size}$
- Imitators
 - Only come to your store
 - If $\text{WTP} > \text{Price}$ then they buy
 - Number of imitators coming to your store depends on how much of the market you have served
 - Daily Arrivals = $q * \text{remaining market size} * \% \text{ of the market served}$
- Advertising
 - You can pay to bring people to your store
 - If $\text{WTP} > \text{Price}$ then they buy
 - Daily Arrivals = $p * \text{remaining market size} * \text{advertising spend} / \500
 - We recommend you do not advertise today as you don't really have the capacity to keep up with it. Wait until next week when you have a faster factory.

Bass Model for Arrival Rates

Assumptions:

Innovators, $p = 0.0002$

Imitators, $q = 0.0035$

Advertising = \$0

Market size = 34,500

Bass Model for Arrival Rates

Assumptions:

Innovators, $p = 0.0002$

Imitators, $q = 0.0035$

Advertising = \$0

Market size = 34,500

Innovator arrivals = $p * \text{remaining market size}$

Imitator arrivals = $q * \text{remaining market size} * \% \text{ of the market served}$

Day 1:

Innovators arrivals = $0.0002 * 34,500 = 6.90$

Imitator arrivals = $0.0035 * 34,500 * 0 / 34,500 = 0$

Day 2:

Innovators arrivals = $0.0002 * 34,993.10 = 6.8986$

Imitator arrivals = $0.0035 * 34,993.10 * 6.90 / 34,500 = 0.0241$

Bass Model Spreadsheet

No Advertising

Enter inputs only in yellow cells. If you don't enter a period end day, the model calculates up to day 2,930. If you don't enter another type of input for a future period, the model uses the value from the latest period with an input.

Market Name	MD - Heart - Temporal	
Market research estimate of initial market size in a regular region: 20,000-40,000. Market research estimate of initial willingness to pay (WTP) range: \$600-\$865. Every \$500 spent on advertising generates arrivals = 0.0002 * the remaining market. Innovator arrivals = 0.0002 * the remaining market. Imitators arrivals = 0.0035 * the remaining market * the share of the initial market that has purchased your product.		
Initial Market Size	34,500	
Mean of WTP	\$723	
Standard Deviation of WTP	\$30	
Approx. WTP Range (mean ± 3s)	\$633	\$813
Begin Forecast On Game Day #	19	
Historic Retail Units Sold	0	
Historic Retail Revenue	\$0	
Historic COGS	\$0	
Historic Retail Fulfillment	\$0	
End Forecast on Game Day #	2,930	
Quarter End Days		
Q1: 91	Q5: 455	Q9: 819
Q2: 182	Q6: 546	Q10: 910
Q3: 273	Q7: 637	Q11: 1001
Q4: 364	Q8: 728	Q12: 1092
Q13: 1183	Q14: 1274	Q15: 1365
Q16: 1456	Q17: 1517	Q18: 1618
Q19: 1719	Q20: 1819	

[illegible]

Bass Model Spreadsheet

No Advertising

Daily Calculations

	Adv. (p / \$500)	Innovation (p)	Imitation (q)
Coefficient	0.0002	0.0002	0.0035

Notes: BOP = Beginning of Period; EOP = End of Period

Total sales cannot be greater than either total arrivals or the remaining market size.

Search
by Day:

Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ([A+A _t]/M)
19	\$ 700.00	\$ -	-	6.90	-	6.90	77.94%	-	5.38	-	5.38	-	0.00%	34,500	5	0.02%
364	\$ 700.00	\$ -	-	6.33	9.18	15.51	75.94%	-	4.81	6.97	11.78	2,861	8.29%	31,639	2,872	8.33%
728	\$ 700.00	\$ -	-	5.17	22.70	27.87	70.53%	-	3.65	16.01	19.66	8,661	25.10%	25,839	8,681	25.16%
1,092	\$ 700.00	\$ -	-	3.67	30.07	33.74	58.43%	-	2.14	17.57	19.71	16,171	46.87%	18,329	16,190	46.93%

Quarter	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ([A+A _t]/M)
1	19	\$ 700.00	\$ -	-	6.90	-	6.90	77.94%	-	5.38	-	5.38	-	0.00%	34,500	5	0.02%
1	20	\$ 700.00	\$ -	-	6.90	0.02	6.92	77.94%	-	5.38	0.01	5.39	5	0.02%	34,495	11	0.03%
1	21	\$ 700.00	\$ -	-	6.90	0.04	6.94	77.93%	-	5.38	0.03	5.41	11	0.03%	34,489	16	0.05%
1	22	\$ 700.00	\$ -	-	6.90	0.06	6.95	77.93%	-	5.37	0.04	5.42	16	0.05%	34,484	22	0.06%
1	23	\$ 700.00	\$ -	-	6.90	0.08	6.97	77.93%	-	5.37	0.06	5.43	22	0.06%	34,478	27	0.08%
1	24	\$ 700.00	\$ -	-	6.89	0.09	6.99	77.92%	-	5.37	0.07	5.45	27	0.08%	34,473	32	0.09%
1	25	\$ 700.00	\$ -	-	6.89	0.11	7.01	77.92%	-	5.37	0.09	5.46	32	0.09%	34,468	38	0.11%
1	26	\$ 700.00	\$ -	-	6.89	0.13	7.03	77.92%	-	5.37	0.10	5.47	38	0.11%	34,462	43	0.13%
1	27	\$ 700.00	\$ -	-	6.89	0.15	7.04	77.91%	-	5.37	0.12	5.49	43	0.13%	34,457	49	0.14%
1	28	\$ 700.00	\$ -	-	6.89	0.17	7.06	77.91%	-	5.37	0.13	5.50	49	0.14%	34,451	54	0.16%
1	29	\$ 700.00	\$ -	-	6.89	0.19	7.08	77.91%	-	5.37	0.15	5.52	54	0.16%	34,446	60	0.17%
1	30	\$ 700.00	\$ -	-	6.89	0.21	7.10	77.90%	-	5.37	0.16	5.53	60	0.17%	34,440	65	0.19%
1	31	\$ 700.00	\$ -	-	6.89	0.23	7.12	77.90%	-	5.36	0.18	5.54	65	0.19%	34,435	71	0.21%
1	32	\$ 700.00	\$ -	-	6.89	0.25	7.13	77.90%	-	5.36	0.19	5.56	71	0.21%	34,429	77	0.22%
1	33	\$ 700.00	\$ -	-	6.88	0.27	7.15	77.89%	-	5.36	0.21	5.57	77	0.22%	34,423	82	0.24%
1	34	\$ 700.00	\$ -	-	6.88	0.29	7.17	77.89%	-	5.36	0.22	5.58	82	0.24%	34,418	88	0.25%
1	35	\$ 700.00	\$ -	-	6.88	0.31	7.19	77.88%	-	5.36	0.24	5.60	88	0.25%	34,412	93	0.27%
1	36	\$ 700.00	\$ -	-	6.88	0.33	7.21	77.88%	-	5.36	0.25	5.61	93	0.27%	34,407	99	0.29%
1	37	\$ 700.00	\$ -	-	6.88	0.35	7.23	77.88%	-	5.36	0.27	5.63	99	0.29%	34,401	105	0.30%
1	38	\$ 700.00	\$ -	-	6.88	0.36	7.24	77.87%	-	5.36	0.28	5.64	105	0.30%	34,395	110	0.32%

Bass Model Spreadsheet

\$500 Advertising

Enter inputs only in yellow cells. If you don't enter a period end day, the model calculates up to day 2,930. If you don't enter another type of input for a future period, the model uses the value from the latest period with an input.

Market Name	MD - Heart - Temporal	Initial Market Size	34,500	Historic Retail Units Sold	0	Quarter End Days									
Market research estimate of initial market size in a regular region: 20,000-40,000.						Q1: 91 Q5: 455 Q9: 819 Q13: 1183 Q17: 15									
Market research estimate of initial willingness to pay (WTP) range: \$600-\$865.						Q2: 182 Q6: 546 Q10: 910 Q14: 1274 Q18: 16									
Every \$500 spent on advertising generates arrivals = 0.0002 * the remaining market.						Q3: 273 Q7: 637 Q11: 1001 Q15: 1365 Q19: 17									
Innovator arrivals = 0.0002 * the remaining market.						Q4: 364 Q8: 728 Q12: 1092 Q16: 1456 Q20: 18									
Imitators arrivals = 0.0035 * the remaining market * the share of the initial market that has purchased your product.															
		Standard Deviation of WTP	\$30			Historic COGS		\$0							
		Approx. WTP Range (mean ± 3s)	\$633	\$813			Historic Retail Fulfillment		\$0						
		Begin Forecast On Game Day #	19			End Forecast on Game Day #		2,930							

[illegible]

Bass Model Spreadsheet

\$500 Advertising

Daily Calculations

	Adv. (p / \$500)	Innovation (p)	Imitation (q)
Coefficient	0.0002	0.0002	0.0035

Notes: BOP = Beginning of Period; EOP = End of Period

Total sales cannot be greater than either total arrivals or the remaining market size.

Search by Day:	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
	19	\$ 700.00	\$ 500.00	6.90	6.90	-	13.80	77.94%	5.38	5.38	-	10.76	-	0.00%	34,500	11	0.03%
	364	\$ 700.00	\$ 500.00	5.83	5.83	15.80	27.47	73.89%	4.31	4.31	11.68	20.30	5,343	15.49%	29,157	5,363	15.55%
	728	\$ 700.00	\$ 500.00	4.15	4.15	28.94	37.24	63.31%	2.63	2.63	18.32	23.58	13,741	39.83%	20,759	13,765	39.90%
	1,092	\$ 700.00	\$ 500.00	2.71	2.71	28.79	34.21	43.69%	1.18	1.18	12.58	14.95	20,956	60.74%	13,544	20,971	60.78%

Quarter	Day (t)	Retail Price per Unit	\$ of Advertising	Advertising Arrivals ($Q_{a,t}$)	Innovator Arrivals ($Q_{p,t}$)	Imitator Arrivals ($Q_{q,t}$)	Total Arrivals (Q_t)	Sales Rate	Advertising Sales ($A_{a,t}$)	Innovator Sales ($A_{p,t}$)	Imitator Sales ($A_{q,t}$)	Total Sales (A_t)	BOP Cumulative Sales (A)	BOP Market Penetration % (A/M)	BOP Remaining Market Size (M-A)	EOP Cumulative Sales (A+A _t)	EOP Market Penetration % ($[A+A_t]/M$)
1	19	\$ 700.00	\$ 500.00	6.90	6.90	-	13.80	77.94%	5.38	5.38	-	10.76	-	0.00%	34,500	11	0.03%
1	20	\$ 700.00	\$ 500.00	6.90	6.90	0.04	13.83	77.93%	5.38	5.38	0.03	10.78	11	0.03%	34,489	22	0.06%
1	21	\$ 700.00	\$ 500.00	6.90	6.90	0.08	13.87	77.93%	5.37	5.37	0.06	10.81	22	0.06%	34,478	32	0.09%
1	22	\$ 700.00	\$ 500.00	6.89	6.89	0.11	13.90	77.92%	5.37	5.37	0.09	10.83	32	0.09%	34,468	43	0.13%
1	23	\$ 700.00	\$ 500.00	6.89	6.89	0.15	13.93	77.91%	5.37	5.37	0.12	10.86	43	0.13%	34,457	54	0.16%
1	24	\$ 700.00	\$ 500.00	6.89	6.89	0.19	13.97	77.91%	5.37	5.37	0.15	10.88	54	0.16%	34,446	65	0.19%
1	25	\$ 700.00	\$ 500.00	6.89	6.89	0.23	14.00	77.90%	5.36	5.36	0.18	10.91	65	0.19%	34,435	76	0.22%
1	26	\$ 700.00	\$ 500.00	6.88	6.88	0.26	14.03	77.89%	5.36	5.36	0.21	10.93	76	0.22%	34,424	87	0.25%
1	27	\$ 700.00	\$ 500.00	6.88	6.88	0.30	14.07	77.89%	5.36	5.36	0.24	10.96	87	0.25%	34,413	98	0.28%
1	28	\$ 700.00	\$ 500.00	6.88	6.88	0.34	14.10	77.88%	5.36	5.36	0.27	10.98	98	0.28%	34,402	109	0.32%
1	29	\$ 700.00	\$ 500.00	6.88	6.88	0.38	14.14	77.87%	5.36	5.36	0.30	11.01	109	0.32%	34,391	120	0.35%
1	30	\$ 700.00	\$ 500.00	6.88	6.88	0.42	14.17	77.86%	5.35	5.35	0.33	11.03	120	0.35%	34,380	131	0.38%
1	31	\$ 700.00	\$ 500.00	6.87	6.87	0.46	14.20	77.86%	5.35	5.35	0.35	11.06	131	0.38%	34,369	142	0.41%
1	32	\$ 700.00	\$ 500.00	6.87	6.87	0.49	14.24	77.85%	5.35	5.35	0.38	11.08	142	0.41%	34,358	153	0.44%
1	33	\$ 700.00	\$ 500.00	6.87	6.87	0.53	14.27	77.84%	5.35	5.35	0.41	11.11	153	0.44%	34,347	164	0.48%
1	34	\$ 700.00	\$ 500.00	6.87	6.87	0.57	14.31	77.84%	5.35	5.35	0.44	11.13	164	0.48%	34,336	175	0.51%
1	35	\$ 700.00	\$ 500.00	6.86	6.86	0.61	14.34	77.83%	5.34	5.34	0.47	11.16	175	0.51%	34,325	186	0.54%
1	36	\$ 700.00	\$ 500.00	6.86	6.86	0.65	14.37	77.82%	5.34	5.34	0.50	11.19	186	0.54%	34,314	197	0.57%
1	37	\$ 700.00	\$ 500.00	6.86	6.86	0.69	14.41	77.81%	5.34	5.34	0.53	11.21	197	0.57%	34,303	209	0.60%
1	38	\$ 700.00	\$ 500.00	6.86	6.86	0.73	14.44	77.81%	5.34	5.34	0.56	11.24	209	0.60%	34,291	220	0.64%

Back to Basics

SUPPLY = DEMAND

Financing

How will we finance our investing and operating activities?

- Retained Earnings
- Working Capital
- Borrow
 - 5-year zero coupon bonds
 - Excellent, 10% APR, 20x interest coverage ratio
 - Good, 15% APR, 7x interest coverage ratio
 - Poor, 25% APR, 2x interest coverage ratio
 - Daily compounding

When can we borrow?

- Need a full quarter of positive EBIT (operating income)
- Borrowing based on your interest coverage ratio
- Interest coverage ratio = $(4 * \text{quarterly EBIT}) / (\text{interest expense if you borrow})$
- 5-year zero coupon bonds
- Daily compounding
- Effective annual rate = $(1 + \text{APR}/n)^n - 1$

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at excellent?

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at good?

Issuing Bonds Example

EBIT Q5 = \$25,000

How much can we borrow at poor?

Debt Model

\$25,000 in Quarterly EBIT

Inputs

What is your current quarterly interest expense?	\$0	<i>per quarter</i>	\$0	<i>per year</i>
What was your operating income for the last full quarter?	\$25,000	<i>per quarter</i>	\$100,000	<i>per year</i>

Credit Rating Criteria and Interest Rates

	Excellent	Good	Poor
Interest Coverage Hurdle for Credit Rating	20x	7x	2x
Annual Percentage Rate (APR)	10.00%	15.00%	25.00%
Effective Annual Rate (EAR)	10.516%	16.180%	28.392%

Maximum Interest Expense to Retain Credit Rating

	Excellent	Good	Poor	Total
Cumulative Annualized Interest Expense	\$5,000.00	\$14,285.71	\$50,000.00	\$50,000.00
Incremental Annualized Interest Expense	\$5,000.00	\$9,285.71	\$35,714.29	\$50,000.00

Debt Capacity

	Excellent	Good	Poor	Total
# of Issuable Bonds	78	121	439	639
Face Value of Issuable Bonds	\$78,388.88	\$121,477.27	\$438,869.13	\$638,735.27
Present Value of Issuable Bonds (Cash Received)	\$47,548.52	\$57,390.66	\$125,792.07	\$230,731.26

Value of a Single Bond

	Excellent	Good	Poor
Face Value of One Bond	\$1,000.00	\$1,000.00	\$1,000.00
Present Value of One Bond	\$606.57	\$472.44	\$286.63

Issuing Bonds Example Solutions

EBIT Q5 = \$25,000

How much interest expense can we support at the Excellent rate?	$\$100,000 / \text{interest expense} = 20$ interest expense = \$5,000
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.10 / 364)^{364} - 1 = 0.10516$ $\$5,000 = PV * 0.10516$ PV = \$47,548.52
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$47,548.52 = FV / (1 + 0.10516)^5$ FV = \$79,388.88
How much interest expense can we support at the Good rate?	$\$100,000 / 7 = \$14,285.71$
Subtract the interest expense already allocated to the Excellent debt. This is how much interest expense can be allocated to the Good debt.	$\$14,285.71 - \$5,000 = \$9,285.71$
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.15 / 364)^{364} - 1 = 0.1618$ $\$9,285.71 = PV * 0.1618$ PV = \$57,390.66
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$57,390.66 = FV / (1.1618)^5$ FV = \$121,477.27
How much interest expense can we support at the Poor rate?	$\$100,000 / 2 = \$50,000$
Subtract the interest expense already allocated to the Excellent and Good debt. This is how much interest expense can be allocated to the Poor debt.	$\$50,000 - \$14,285.71 = \$35,714.29$
What is the present value of the debt that would generate the interest expense above?	$r = (1 + 0.25 / 364)^{364} - 1 = 0.283915$ $\$35,714.29 = PV * 0.283915$ PV = \$125,792.07
What is the future value (face value) of debt that corresponds to the present value of the debt calculated above?	$\$125,792.07 = FV / (1.283915)^5$ FV = \$438,869.13

Borrowing

Headquarters – you can't borrow when you start as you have negative EBIT

BarterTown Headquarters



Hover over values for details. Click on numbers and icons to open panels and edit values.

As of day 364 ↺

Finance

Operating Income : \$-105,160 (Q 4)

Cash : \$1,579,530

Working capital ratio : 18.08

Trade credit : materials net / 30
others net / 15

Bonds : \$0 0 issuable

Dividends : \$0

Marketing

Designs : 0 ... 0  1  0 

Retail sales : 1 of 102 regional markets
1 inventories
\$0 per day advertising
2,837 units sold total
\$1,985,900 total revenues

Sales to other teams : 0 inventories
0 units sold total
\$0 total revenues

Operations

Facilities :  

All agreements : 0 ... 1  0 

263 units on hand
0 units in transit

Production : 0 ... 1  0 

100 units in process

Procurement : 0 ... 0  0 

Fulfillment : 0 ... 0  0 

Long-Term Debt Panel for Bonds

Long-term debt

Up to 1,680 bonds issuable at Excellent (10% YTM)

0 Bond Issue     **ADD NEW +**

10 records Search:

Day	Terms	Count			Value	YTM	Days to maturity	Redemption price
		Total	Outstanding	Redeemed				
					Face			

No data available in table

Long-term debt

Up to 2,584 bonds issuable at Good (15% YTM)

1 Bond Issue     **ADD NEW +**

10 records Search:

Day	Terms	Count			Value	YTM	Days to maturity	Redemption price
		Total	Outstanding	Redeemed				
					Face			
838	5 zero cpn 	1,680	1,680	0 	\$1,680,000	10%	1,819	\$614.07

Showing 1 to 1 of 1 entries  1 

Bond Terms Panel : BarterTown

As of day 830.12 

Rating : Excellent
Issue price : \$ 613.91
Maturity : 5years 
Issue count : 1680 
Par : \$1,000
Yield to maturity : 10% per annum
Coupon rate : 0% 
 Callable at accreted value starting immediately

Notes

Practice Game

Important Links and Documents

See Class 3 on Canvas

- Practice Game Link
- Important Documents
 - Quick Reference for Practice Game
 - Market Research for Practice Game
 - Gleacher Game Case Brief
 - Product Design Guide
 - Suggested Team Roles
 - Player Manual
- Models (all complete, with formulas, adapt to your needs)
 - Production Model
 - Bass Model
 - Debt Model

Chat is in Envelopes

Envelope turns yellow when there's a new message in a closed chat window

- The group chat icon  is to the right of game status above the map.
 - Use this to contact all teams during a game.
- The admin chat icon  is to the right of group chat.
 - Use this to contact just Kathleen or the TAs during a game.
- If you want to send a message to a specific team, click on the envelope  in their home region on the map.
- If there is a new message, the chat icon turns yellow.
 - The icon will not turn yellow if the chat window is already open.



Practice Game

Play

- Start at the end of quarter 4 (day 365) and end on day 1092
- Everyone starts at the same point, which is where we started yesterday.
- Everything you learned yesterday applies today.
- Play 3 quarters in class, from 4:45pm-6:15pm
- Play 3 more quarters tonight, from 7pm-8:30pm, ending live play on day 910.
- Speed: 2 quarters per hour, 30 minutes per quarter, 3 game days per minute.
- We will run the game quickly through day 1092, so you can see what would have happened if your company ran two more quarters.
- **Please do the homework together – it is important to all be on the same page with your strategy. The homework will take approximately 1.5-2 hours. It is the last long assignment this week.**

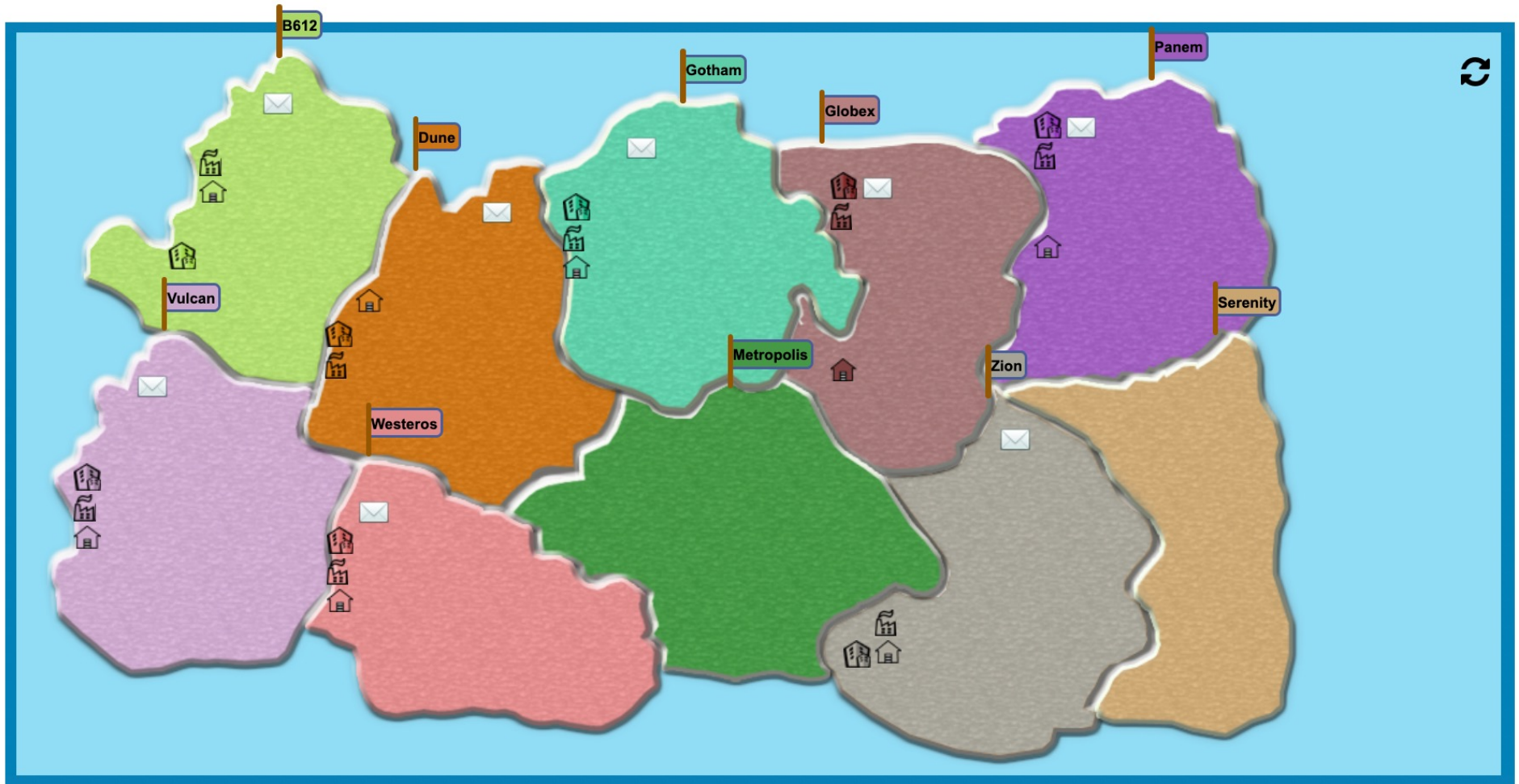
Things to Consider

Use the Quick Reference
Guide!

- Market sizes are the same in all regions, with the exception of Metropolis
- Willingness to pay is the same in all regions
- You can trade with all teams
- You can change your batch sizes
- You can speed up or slow down your factory
 - Through CAPEX
 - Through Daily Expenditure
- You can use containers or mail for shipping
 - Consider time
 - Consider how many units you need in a container to equal the per unit mailing cost
- You can ship to wholesale customers from the factory or from your DC
- You can only sell to retail customers through the DCs
- Retailers pay handling and commission

Practice Game Map

Serenity (military) and Metropolis (large) have no home teams



Starting Point

World	Team Name	Return to Investors	Retained Earnings at end of day 364	Cash Position
	B612	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Dune	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Globex	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Gotham	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Panem	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Vulcan	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Westeros	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>
	Zion	\$ <u>1,579,530</u>	\$ -342,398	\$ <u>1,579,530</u>

Practice Game – Big Picture

- Start with one product
 - Heart View
- Build a new factory to help become profitable
 - You have \$1,549,077 now – extra cash for practice
 - Use Production Model
- Design, build and sell products
 - Advertise
 - Use Bass Model
- Borrow money
 - Use Debt Model
- Expand your business into new regions
 - Borrow and build a DC
 - Make partnerships

Suggested Steps

- Build a new line factory and **give it a name**
 - Schedule the old one to close shortly after the new one opens
 - Clone your shipping agreements
- Build a new product or two and **give them names**
 - Make sure they can be profitable in your line factory
 - Design, develop, price
 - Set up a shipping agreement with your new factory to produce it
 - Set up shipping agreements with competitors to sell it
 - Don't forget to make your re-order points > 0
- Focus on running a profitable business
 - As soon as you have a profitable quarter, you can borrow cash
- Borrow and expand
 - Products
 - Distribution Center
 - Factories

Markets for Today

See the Market Research Report for Practice Game for details

- Athlete
- MD Heart
- MD Dissolved Gasses
- MD Cancer (Breast)
- MD Cancer (Bladder & Kidney)
- MD Fertility (Estrogen)
- MD Metabolic (Bilirubin)
- Law (Narcotic)
- Clinical Fertility
- Clinical Cardiovascular
- Military (Botulinum)
- Military (Anatoxin)

Use the Reports

See the Player Manual for details

- Access reports through the HQ
 - Financial Statements
 - Working Capital Analysis
 - Product Design
 - Product Line Architecture
 - Regional Markets
 - Market Grid
 - Focus Groups
 - Consolidated Inventories
 - Inventories Grid
 - Consolidated Shipping Agreements
- “Open Report” from the map screen
 - Financial Statements
 - Working Capital Analysis
 - Market Grid
 - Product Analysis
 - Price Response
 - Product Line Architecture
 - Product Designs
 - Focus Groups
 - Inventory Status
 - Inventory Grid
 - Active Shipping Agreement Sankey
 - Factories
 - Distribution Centers
 - Active Factories Grid

Things to Remember

Seriously, read these!

- You can build new factories.
- You can build new distribution centers.
- Set up shipping agreements at the same time you start construction on factories and DCs, and at the same you start development of new products.
- Name your factories and products in a way that makes them easy to identify.
- You can speed up or slow down your factory. Never set labor to a very low amount. Set it either to 0 to shut down the factory or to something reasonable for production. Only add incremental capital to a factory.
- Understand why factories go idle and what to do about it. Understand why you stock out and what to do about it. Watch your inventory status report to identify service interruptions and build up of inventory.
- Make sure the products you design or purchase include the features you think they do.
- Make sure you run focus groups for the correct markets.
- Don't sell things for less than what they cost you to produce.
- You can trade with anyone. Agree on terms through chat or in person before sending shipping agreements.
- When sending shipping agreements to other teams, include some WTP information in the comment section to cut down on negotiation time.
- Remember that if you find a market in your region, the same market, WTP and market size exists in all other regions.
- Don't forget that the commission and handling charges are paid by the retailer and shipping is paid by the wholesaler. Incorporate that information in your offers.
- You can use containers or mail for shipping.
- You can ship to wholesale customers from your DC or your factory.
- You can only sell to retail customers through the DCs.
- Make sure you understand how priorities and reorder points work. Make your reorder points > 0 for active shipping agreements. To suspend a shipping agreement, put the re-order point at -1 .

Things to Remember

Seriously, read these!

- Use the scheduling features to shut down factories, start and stop advertising, schedule construction projects, start development, among other things.
- Review your financials frequently to catch problems.
- Make sure the left hand knows what the right hand is doing.
- Choose strategic partners wisely.
- Do not use outdated course materials; only use those for today.
- Use the Quick Reference guide.
- Use the Player Manual.
- Use the reports.
- Use your models. We made them for a reason.
- Watch the chat panel as we put tips and answer questions there, and other teams may be trying to contact you.

For More Tips

CHECK CHAT

Back to Basics

SUPPLY = DEMAND

Objective

MAXIMIZE CURRENT SHAREHOLDER VALUE

(Your company is a going concern – operate for the long run)

* WARNING – REAL GAME STARTING POINT *

- You will only have \$549K in cash when we start the real game on Wednesday
- After changing container to mail – you will need to buy a production line, but you don't have \$600,000.
- You will need a PLAN to get to \$600K.
- You should NOT run focus groups or build any new products until you have the production line under construction.
- Please run numbers and have a concrete plan to get to \$600K. This is part of Assignment 4.
- You have all the data you need, so put your analytical minds to work.

Good Luck!

Appendix: Possible Roles

CEO

- Responsible for overall success and strategy of the business for the foreseeable future.
- Develop a strategy that positions the company to maintain a competitive advantage beyond the duration of the simulation.
- Lead the senior management team.
- Facilitate cooperation and communication among the leaders of various functions or business units.
- Anticipate changes in the competitive landscape by identifying and tracking key performance indicators.
- Work with CMO and CSO to identify when to enter and exit markets.
- Work with the CFO and COO to make capital allocation decisions.
- Work with the CFO to determine your growth and dividend payout strategy.
- Work closely with the leadership team to build the strategy for the period following the simulation and manage the final simulation write-up.

CFO

- Manage cash flows from operating, investing and financing activities.
- Prepare financial projections and budgets for business.
- Manage working capital through trade credit terms, DSO and other tools at your disposal.
- Manage the debt and credit ratings.
- Work with the COO to determine the appropriate transfer price from factory to distribution center.
- Work with the COO to develop appropriate cost measures to support pricing and production decisions.
- Work with the CEO and the rest of the management team to determine the financial impact and trade-off of possible product, pricing and capacity building strategies.
- Financial statement analysis and preparation of profitability summary reports, re: products, distribution centers, factories and the business as a whole.
- Work with marketing, production and business development as needed to help them estimate the financial impacts of their decisions.

COO

- Design a supply chain that delivers a sustainable advantage.
- Manage inventory throughout the supply chain.
- Monitor inventory levels and make sure that inventory is in the right place at the right time.
- Manage shipping agreements and all of its attributes, such as batch sizes, shipping mechanisms, and reorder points.
- Anticipate inventory buildup and stock outs and advise management team accordingly.
- Optimize production capabilities for given production technologies and work in conjunction with the marketing and financial leaders of the organization to make choices about capacity building decisions.
- Optimize distribution.
- Calculate unit production costs for products using different technologies and levels of capital expenditures and recurring costs.
- Provide marketing team with cost forecasts aligned with their demand forecasts.
- Determine the financial and strategic trade-offs of building for efficiency, flexibility, fast response, etc.
- Manage short term, medium term and long capacity through order quantities, recurring expenditure levels, and capital expenditures.
- Work with the CFO to keep track of profitability at the factory and distribution center levels.
- Track and manage idle time in the factories

CMO

- Develop pricing strategies.
- Estimate demand and price elasticity by performing market research and analyzing sales data for your firm and competitor firms.
- Work with the CSO to research competitors' products and prices.
- Design new products and manage the new product pipeline.
- Set sales volume targets and prices for new and old products.
- Work closely with the production team to keep them informed of changes in demand and new product development and understand when inventory is building up and determine whether it needs to be marked down.
- Work with the CFO to evaluate the financial aspects of product development.
- Manage product portfolio.

CSO

- Business Development
- Work with the CEO and CMO to develop distribution strategies.
- Work with the CFO to determine contract provisions and terms and execute the contracts.
- Negotiate contracts with other teams to manage product distribution and establish distribution channels.
- Monitor what other teams are doing in terms of strategies, products, investments and report back to the rest of the management team.